「Def. Let $\beta: I \to \mathbb{R}$ be a unit-speed curve.
A vector field β' is called the unit tangent vector field on β, denote $T = \beta'$.
The derivative $T' = \beta''$ is called the curvature vector field of β.
The value $\kappa(s) = \|T'(s)\|$ $(s \in I)$ is called the curvature function of β.」

To restriction s.t $\kappa > 0$,
the unit-vector field $N = \frac{T'}{\kappa}$ is called the principal normal vector field of β.
The cross product $B = T \times N$ on β is called the binormal vector field of β.

「Lemma. Let β be a unit-speed curve in $\mathbb{R}^3$ with $\kappa > 0$,
Then, the three vector fields T, N, and B on β are unit vector fields that are mutually orthogonal at each point.
Proof. By definition, $\|T\| = \|\beta'\| = 1$. And since the curvature $\kappa > 0$, $\|N\| = \frac{1}{\kappa} \cdot \|T'\| = 1$.
And, since $\|T\|^2 = T \cdot T$ is constant, $2T \cdot T' = 0$, so $T \perp N$.
therefore the binormal, $\|B\| = \|T \times N\|^2 = \|T\|^2 \cdot \|N\|^2 - (T \cdot N)^2 = 1 - 0 = 1$. ∎

In summary, $T = \beta'$, $N = T'/\kappa$, $B = T \times N$, and $\{T, N, B\}$ is orthonormal.
Since $B \cdot T = 0$, $B' \cdot T + B \cdot T' = 0$ by differentiate. Therefore,

$$B' \cdot T = -B \cdot T' = -B \cdot \kappa N = 0$$

And since B is unit, $B' \cdot B = 0$, thus, there is a scalar value function τ s.t

$$B' = -\tau N$$

, called the torsion. Now

「Thm. Let β be a unit-speed curve in $\mathbb{R}^3$ with curvature $\kappa > 0$ and torsion τ,

$$\begin{aligned} T' &= \kappa N \\ N' &= -\kappa T + \tau B \\ B' &= -\tau N \end{aligned} = \begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix} \begin{bmatrix} T \\ N \\ B \end{bmatrix}$$

Proof. The first and third rows are just definition. To show the second, observe
$N' = (N' \cdot T)T + (N' \cdot N)N + (N' \cdot B)B$
Now, $N' \cdot T = N \cdot T'$ and $N' \cdot B = N \cdot B'$ gives the results.

Prop. Let β be a unit-speed curve in $\mathbb{R}^3$ with $\kappa > 0$. Then,

β is a plane curve if and only if $\tau = 0$, the torsion is zero.

Proof. Suppose that β is a plane curve. That, $\mathrm{Im}\,\beta$ lies in some plane.
Therefore, there are two vectors $p, q \in \mathbb{R}^3$ such that $q \neq 0$ and

$$[\beta(s) - p] \cdot q = 0 \quad \text{for all } s.$$

Differentiating yields:

$$\beta'(s) \cdot q = 0 \quad \text{and} \quad \beta''(s) \cdot q = 0 \quad \text{for all } s.$$

That is, q is orthogonal to $T = \beta'$ and $N = \beta''/\kappa$. So, B and q are parallel,
we can conclude that $B = \pm q/\|q\|$. So $B' = 0$, the torsion τ must be zero.
Conversely, suppose that the torsion is zero. Then, $B' = 0$ by definition.
($B' = 0$ means B is parallel.) To show that $\mathrm{Im}\,\beta$ lies in some plane, define

$$f(s) = [\beta(s) - \beta(0)] \cdot B(s) \text{ for all } s.$$

Then, the derivative is

$$f'(s) = \beta'(s) \cdot B(s) + \beta(s) \cdot B'(s) = \beta'(s) \cdot B(s) \overset{\text{def.}}{=} T \cdot B = 0.$$

By definition, $f(0) = 0$, so $f = 0$. Consequently,

$$[\beta(s) - \beta(0)] \cdot B = 0 \quad \text{for all } s,$$

So proved ☒

Prop. Let β be a unit speed curve with constant curvature $\kappa > 0$ and torsion zero.
Then, the β is part of a circle of radius $1/\kappa$.
Proof. Since $\tau = 0$, the β is a plane curve. Define a curve

$$\gamma(s) = \beta(s) + \frac{1}{\kappa} \cdot N(s).$$

By the Frenet formular, $\gamma' = \beta' + \frac{1}{\kappa} \cdot N' = T + \frac{1}{\kappa} \cdot (-\kappa T) = 0$. Therefore, γ is constant.
Now, $d(\gamma, \beta(s)) = \|\gamma - \beta(s)\| = \frac{1}{\kappa} \cdot \|N\| = \frac{1}{\kappa}$, so proved.